A BOUNDED ABOVE COMPLEX WHOSE TERMS ARE SHIFT-RELATIVELY-PSEUDO-COHERENT IS RELATIVELY PSEUDO-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17082-17089 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17091-17095, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Term version of this wave's L16194 cohomology version; both feed the relative-pseudo-coherence toolkit opened in waves 12-15.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohomology, lemma cohomology-lemma-complex-pseudo-coherent-modules; the chart definition of relative pseudo-coherence.

Source reference. Original lemma 09V7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17095–17095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes which is locally of finite type. Let $m \in \mathbf{Z}$. Let $\mathcal{F}^\bullet$ be a (locally) bounded above complex of $\mathcal{O}_X$-modules such that $\mathcal{F}^i$ is $(m-i)$-pseudo-coherent relative to S for all i. Then $\mathcal{F}^\bullet$ is m-pseudo-coherent relative to S.*

Proof. Follows from Cohomology, Lemma 09V7 and the definitions. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.